

Worksheet: Chemical equilibrium in the Haber Process

Name(s) _____

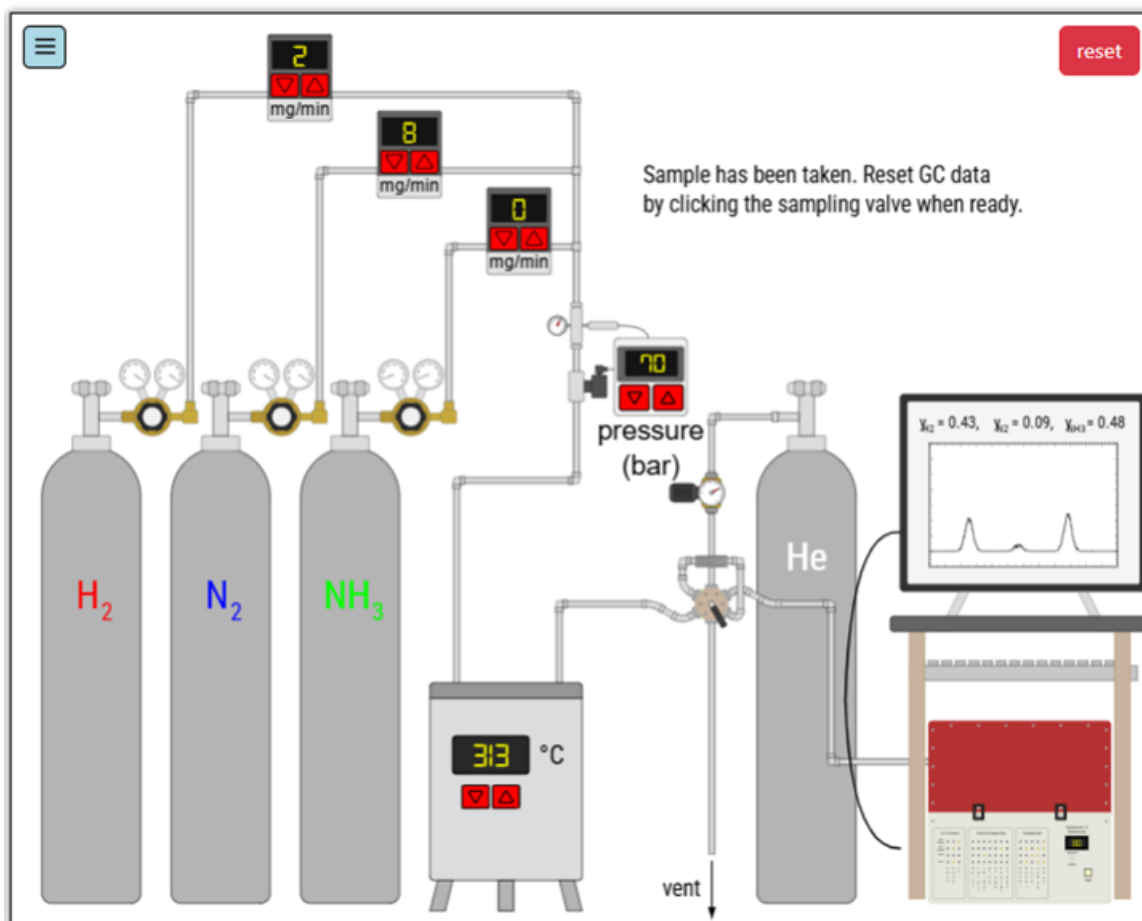
This experiment calculates chemical equilibrium constants for the gas-phase Haber reaction ($\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$) from measurements of equilibrium concentrations. The heat of reaction is determined by measuring equilibrium constants over a range of temperatures.

Student learning objectives

1. Calculate equilibrium constants from equilibrium concentrations.
2. Apply the van't Hoff equation to determine the heat of reaction from equilibrium data.
3. Predict and interpret the effect of pressure, temperature, and feed composition on ammonia equilibrium conversion.

Equipment

This catalytic reaction takes place at high pressures and elevated temperatures. Hydrogen and nitrogen are fed to the reactor to form ammonia. Ammonia can also be added to the reactor feed since in commercial reactors, unreacted N_2 and H_2 (and a fraction of the NH_3 product) recycle back to the feed. A gas chromatograph measures the composition of the reactor outlet stream.



Questions to answer before starting the experiment

1. For an exothermic reaction, does the equilibrium constant increase or decrease as temperature increases? Explain using Le Châtelier's principle and the van't Hoff equation.
2. For the Haber reaction, does increasing pressure favor reactants or products? Explain.
3. At a fixed temperature, how does the equilibrium constant change as pressure increases for the Haber reaction?
4. What are the units of an equilibrium constant?
5. This experiment assumes ideal gases. Why might this assumption be questionable?

Notation

T = temperature (K)

P = pressure (bar)

y_i = mole fraction of component i

P_i = partial pressure of component i (bar)

P^0 = standard state pressure (1 bar)

K_{eq} = equilibrium constant

ΔH_{rxn}^0 = heat of reaction (kJ/mol)

R = ideal gas constant (kJ/(mol K))

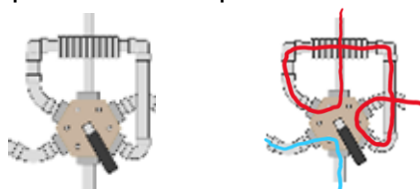
Experimental procedure

Experiment #1 Effect of pressure on equilibrium composition

1. Set the temperature of the sand bath heater to around 350°C by clicking on the up or down arrows. Record the temperature in Celcius and kelvin in Table 1.
2. Click the valves on top of the H₂ and N₂ tanks to open the valves.
3. Click on the valve on top of the He tank for the GC to open the valve.
4. Select H₂ and N₂ mass flow rates, using the red arrows on the mass flow controllers, so the H₂/N₂ mass flow rate ratio is 0.2. Record these flow rates in Table 1. Note that a 0.21:1 H₂/N₂ mass ratio is a 3:1 H₂/N₂ molar ratio.
5. Calculate the feed molar flow rates and record these values in Table 1.
6. Use the red arrows on the pressure gauge/controller to select an operating pressure of 50 bar. Record the pressure in Table 1.
7. The display indicates the system is reaching equilibrium. Then the display shows "Sample ready." The valve should be in the position below to fill the sample loop with the reactor effluent. The figure on the right shows the flow paths. Blue represents the reactor effluent flow through the sample loop, and red represents the He flow path.



8. Click on the switching valve to inject a sample into the GC. The valve will then be in the position shown below as He flushes the sample loop contents into the GC, and the reactor effluent flows directly to the vent. The blue represents the reactor effluent path, and the red represents the He path.



10. Record in Table 1 the mole fractions of H₂, N₂, and NH₃ obtained by the GC.
11. Calculate the equilibrium pressures of N₂, H₂, and NH₃ and record in Table 2.
12. Calculate the equilibrium constant and record the value in Table 2:

$$K_{eq} = \frac{\left(\frac{P_{NH_3}}{P^0}\right)^2}{\left(\frac{P_{H_2}}{P^0}\right)^3 \left(\frac{P_{N_2}}{P^0}\right)}$$

In this equation, partial pressures are in units of bar, and the standard state pressure P^0 is 1 bar. The equation is usually written as indicated below, and the equilibrium constant is dimensionless:

$$K_{eq} = \frac{P_{NH_3}^2}{P_{H_2}^3 P_{N_2}}$$

This equation assumes ideal gases, which is not a good assumption at the high pressures used for ammonia formation, but this assumption simplifies the calculations for this experiment.

13. Repeat these measurements at 75 and 100 bar and record the equilibrium mole fractions and partial pressure in Table 1.

Table 1 Effect of pressure on equilibrium

Temperature = _____ °C = _____ K

H₂ mass feed rate _____ N₂ mass feed rate _____

H₂ molar feed rate _____ N₂ molar feed rate _____

		Equilibrium mole fraction			Equilibrium partial pressure (bar)			
Experiment	P (bar)	y_{N_2}	y_{H_2}	y_{NH_3}	P_{N_2}	P_{H_2}	P_{NH_3}	K_{eq}
#1								
#2								
#3								

14. What can you conclude about the effect of pressure on the equilibrium constant and on the equilibrium conversion?
15. Using the feed and outlet compositions, estimate the extent of reaction from N₂ consumption, H₂ consumption, and NH₃ formation. Do the three estimates agree? If not, what might explain the differences?

Experiment #2 Effect of feed composition on equilibrium composition

1. Select the temperature of the sand bath heater to between 325 and 375°C by clicking on the up or down arrows. Record the temperature in Celcius and Kevinn in Table 2.
2. Select H₂ and N₂ mass flow rates, using the red arrows on the mass flow controllers, so the H₂/N₂ mass flow rate ratio is 0.05. Record thee flow rates in Table 2. Note that a 0.21:1 H₂/N₂ mass ratio is a 3:1 H₂/N₂ molar ratio.
3. Calculate the feed molar flow rates and record these values in Table 1.
4. Use the red arrows on the pressure gauge/controller to select an operating pressure of 75 bar. Record the pressure in Table 1.
5. When the display indicates "Sample ready," click on the switching value to inject a sample into the GC. Record in Table 3 the mole fractions of H₂, N₂, and NH₃ obtained by the GC.
6. Calculate the equilibrium pressures of N₂, H₂, and NH₃ and record in Table 3.
7. Calculate the equilibrium constant and record the value in Table 3:

$$K_{eq} = \frac{P_{NH_3}^2}{P_{H_2}^3 P_{N_2}}$$

8. Repeat these measurements for H₂/N₂ mass flow ratios of 0.2 and 0.4 and record the results in Tables 2 and 3.

Table 2 Effect of feed composition

Temperature = _____ °C = _____ K Pressure = _____

	Feed rates			
	Mass (mg/min)		Molar (mol/min)	
Experiment	N ₂	H ₂	N ₂	H ₂
#1				
#2				
#3				

Table 3 Effect of feed composition on equilibrium conditions

Experiment	Equilibrium mole fraction			Equilibrium partial pressure (bar)			K_{eq}
	y_{N_2}	y_{H_2}	y_{NH_3}	P_{N_2}	P_{H_2}	P_{NH_3}	
#1							
#2							
#3							

- Discuss the effect of molar feed ratio (H_2/N_2) on the equilibrium composition.

Experiment #3 Effect of temperature on equilibrium constant

- Select the temperature of the sand bath heater to 275°C by clicking on the up or down arrows. Record the temperature in Celcius and kelvin in Table 4.
- Select H_2 and N_2 mass flow rates, using the red arrows on the mass flow controllers, so the H_2/N_2 mass flow rate ratio is 0.2. Calculate the feed molar flow rates and record these values in Table 4.
- Use the red arrows on the pressure gauge/controller to select an operating pressure of 80 bar. Record the pressure in Table 4.
- When the display indicates "Sample ready," click on the switching value to inject a sample into the GC. Record in Table 4 the mole fractions of H_2 , N_2 , and NH_3 obtained by the GC.
- Calculate the equilibrium pressures of N_2 , H_2 , and NH_3 and record in Table 4.
- Calculate the equilibrium constant and record the value in Table 4:

$$K_{eq} = \frac{P_{NH_3}^2}{P_{H_2}^3 P_{N_2}}$$

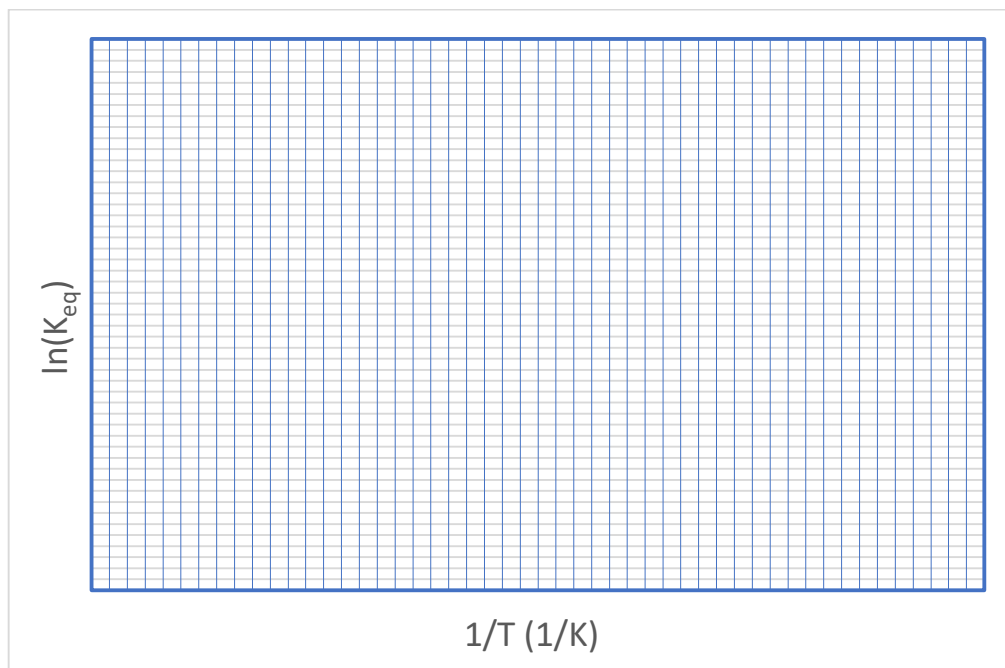
- Repeat these measurements for four more temperatures up to 450°C and record the results in Table 4.

Table 4 Effect of temperature on equilibrium constant

Pressure = _____ Feed molar flow rate ratio (H_2/N_2) _____

Experiment	T(°C)	T (K)	Equilibrium mole fraction			Equilibrium partial pressure (bar)			K_{eq}
			y_{N_2}	y_{H_2}	y_{NH_3}	P_{N_2}	P_{H_2}	P_{NH_3}	
#1									
#2									
#3									
#4									
#5									

8. Plot $\ln(K_{eq})$ versus inverse temperature.



9. Does a straight line fit the data?

10. Calculate the heat of reaction from the van't Hoff equation by linear regression:

$$\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H_{rxn}^0}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

where K_2 (dimensionless) is the equilibrium constant at temperature T_2 .

K_1 (dimensionless) is the equilibrium constant at temperature T_1 .

T_1 and T_2 are absolute temperatures (K)

Note that the slope of the plot of $\ln(K_{eq})$ versus $1/T$ is $-\Delta H_{rxn}^0/R$.

$$\Delta H_{rxn}^0 = \text{_____} \pm \text{_____} \text{ kJ/mol}$$

The heat of reaction from this calculation is for the reaction $N_2 + 3H_2 \rightarrow 2NH_3$. To calculate ΔH_{rxn}^0 per mole of ammonia, divide the above value by 2.

$$\Delta H_{rxn}^0 = \text{_____} \pm \text{_____} \text{ kJ/mol per mole of } NH_3 \text{ formed.}$$

Indicate whether the $\pm$ values are standard deviations or 90% confidence intervals.

11. How does this heat or reaction compare to literature values?

Questions to answer

1. Low temperatures and high pressures favor ammonia synthesis conversion, but commercial reactors operate at elevated temperatures. Why?
2. Why does increasing pressure increase the equilibrium ammonia mole fraction? Explain.
3. At a fixed temperature, did your calculated K_p remain constant as pressure changed? If not, what might explain the difference?
4. Why are catalysts important in the Haber process even though they do not change the equilibrium constant?
5. What assumptions were made in calculating K_p ? Which assumption is most questionable?
6. Where might these measurements have errors?
7. What safety precautions are needed for an actual Haber-process experiment involving H_2 , N_2 , NH_3 , high pressure, high temperature, and a catalyst?